Impact of rice husk derived biochar pyrolysis temperature on food waste anaerobic digestion

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Highlights

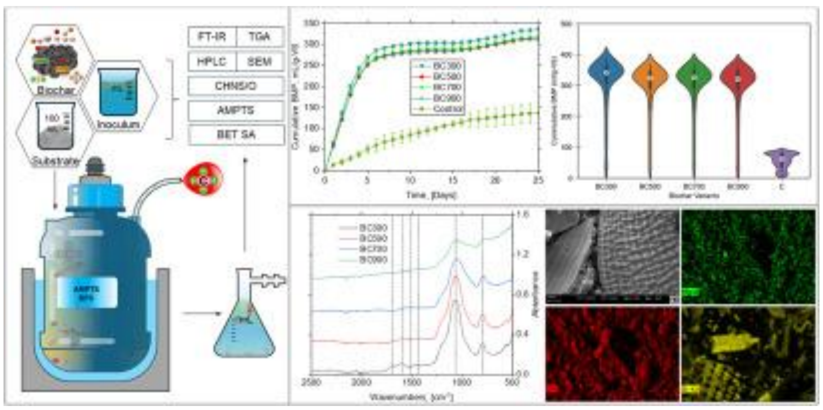
- The anaerobic digestion of food waste was enhanced by rice husk biochar produced at lower temperature.
- The biochar at lower temperature had more porosity favorable for anaerobic digestion.
- The processing temperature did not vary the carbon content of rice husk biochar.

Abstract

This study investigated the influence of rice husk biochar (RHBC) prepared at 300, 500, 700, and 900 °C (BC300, BC500, BC700, and BC900) on the anaerobic digestion (AD) of food waste. The properties of RHBC related to AD, including high heating value, specific surface, porosity, elemental mapping through scanning electron microscopy - energy dispersive X-ray spectroscopy (SEM-EDX), thermal decomposition, electrical conductivity (EC), elemental composition, functional groups, and mcr_A gene copy numbers, were investigated. The specific surface area and mcr_A gene copy numbers are listed as a property of the RHBC relating to AD. Parameters investigated during the AD process were biomethane potential, biomethane production rate, biodegradability, variation in the total volatile fatty acids, specific volatile fatty acids, pH, EC, and chemical oxygen demand, functional groups, changes in the nitrite, nitrate, ammonia, and nitrogen concentration. The study show that increase in the pyrolysis temperature (300 to 900 °C) resulted to the

devolatilization and heteroatom removal in the RHBCs, resulting in decreased concentrations of H, N, and S, while the C content was relatively consistent across pyrolysis temperatures (38.09% to 39.62%). The reactors dopped with BC300 had the highest cumulative BMP at 335.76 mL/g-VS followed by BC500, BC700, and BC900 at 325.63, 316.76, and 311.36 mL/g-VS, respectively. These findings suggest that, compared to highly carbonized biochars made at higher temperatures, lower-temperature biochars, which retain more functional groups and labile components, offer better conditions for the production of biomethane.

Graphical abstract



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Introduction

Food waste (FW) represents a major global challenge, with approximately 1.3 billion tonnes generated worldwide each year [1]. FW is inherently high in moisture content (MC) and highly biodegradable [2]. These characteristics pose challenges in FW management. The food moisture content is 70-80% by mass (wet basis), and the total solid is 20-30% (wet basis), 90% of which are volatile solids (VS) [3]. FW shows variable chemical composition and a high content of biodegradable material (carbohydrates, protein, and lipids) [4]. It is the main component of municipal solid waste (MSW), and it contributes extensively to greenhouse gas (GHG) emissions when landfilled [4]. Several methods are used for its management, including landfilling, incineration, composting and biological treatment processes [5].

Anaerobic digestion (AD) is a notable technology to process FW among many other treatments. It is an economical and established efficient technology to treat biomass for energy recovery and for the production of valuable fertilizer [6,7]. AD is considered a crucial technology in the implementation of the modern circular bioeconomy [8]. To date, numerous studies have focused on optimizing the conditions of AD of FW, including the use of co-digestion strategies, process control, temperature regimes (mesophilic or thermophilic), and the application of various additives aimed at improving process stability and methane yields [9,10]. Nevertheless, AD of FW is still faced with various challenges and inefficient processes due to its high labile organic matter, salt, oil, and protein content, low carbon-to-nitrogen (C/N) ratio, and micronutrient deficiency [4]. FW is composed mostly of easily degradable carbohydrates (50-60% TS), proteins (15-25% TS), and lipids (13-30% TS) and a relatively low C/N ratio [4,11]. It is rich in macro-elements but lacks trace elements. Moreover, FW is generally an acid or sub-acid substrate, which is suitable for biodegradation but sub-optimal for methanogens that operate at slightly higher pH (6.5-7.2) [4].

Therefore, various additives have been investigated to enhance the stability and efficiency of AD of FW, including hydrolytic enzymes, buffering chemicals used to stabilize pH, trace metals supplying essential micronutrients for methanogens, and conductive materials that facilitate direct interspecies electron transfer (DIET). These additives can promote hydrolysis of complex substrates, maintain stable reactor conditions and stimulate microbial metabolism, ultimately improving methane production and overall process efficiency [12].

One of the examples of carbonaceous materials used as AD amendments is biochar (BC). Biochar is an aromatic material derived from biomass pyrolysis [13]. The properties of biochar, including the ability to enhance hydrolysis, immobilization capability, facilitation of DIET, and buffering capacity, significantly influenced the AD process [14]. In addition to pyrolysis temperature, the physicochemical properties of biochar strongly depend on the type of feedstock used for its production. Different biomass precursors lead to biochars with varying mineral composition, surface functional groups and electrical conductivity, which in turn influence microbial colonization and electron transfer processes in anaerobic digesters [12,15]. Moreover, these differences may affect the ability of biochar to adsorb inhibitory compounds and provide conductive surfaces that enhance syntrophic interactions and methane production during AD [12].

Biochars produced from various lignocellulosic residues have been previously investigated as amendments to improve AD performance. For example, hardwood biochar added to the AD of wheat straw doubled methane yield compared with the control, reaching 223 mL CH_4/g-VS at an optimal dosage [16]. Similarly, BC addition during AD of rape straw increased biogas yields by up to 40.5% under mesophilic conditions, indicating that biochar can enhance microbial interactions and methane production in lignocellulosic substrates [17]. Currently, various types of biomass are being investigated as potential feedstocks for biochar production [18].

Among the various biomass sources for biochar production, rice husk has attracted particular attention in rice-producing countries. In the Philippines, rice serves as the staple food and is the second largest crop next to sugarcane, followed by coconut, maize, and banana in terms of production. In 2023, the country produced 20,059,561.96 metric tons of rice and is ranked as the sixth producer in Southeast Asia [19]. Since 2005, the production of rice in the country has increased to 35.29%. On average, the Philippines has an estimated rice husk production of ~2 million tons per year [20,21]. This is equivalent to approximately 5 million barrels of oil equivalent in terms of energy [20]. Converting rice husk into biochars is one of the most effective ways to reuse it [22]. Burning is one of the common methods of rice husk management [22]. Recently, the biochar (BC) production from rice residues has been of increasing interest as it may give sustainable reuse of rice residues for rice-producing countries and their beneficial applications [22]. Notably, RH is characterized by a high silica content, relatively high carbon content and a porous structure after pyrolysis, which may provide favorable conditions for microbial attachment and electron transfer processes in anaerobic digesters [23].

Rice husk is a particularly promising feedstock for biochar production due to its global availability as an agricultural by-product of rice processing. Therefore, in this study, an attempt was made to investigate and further characterize rice husk biochar (RHBC). RHCB has been used as a supplement to improve the AD process. It was demonstrated that adding RHBC during AD systems improves methane production [24,25]. In a study including FW and pig dung, RHBC enhanced methane output by 9.4% [26]. When added to the AD of FW, RHBC pyrolyzed at 550 °C increased methane outputs by up to 36.75% [27]. Currently, the literature does not supply sufficient information on the application of RHBC, especially at a wide range of pyrolysis temperatures (300 to 900 °C). Overall, RHBC pyrolyzed at 500-700 °C, due to its increased C content relative to the raw rice husk, showed an improvement in BMP during the AD process [27,28]. The pyrolysis temperature has a major impact on the characteristics of BC, such as its surface area (SA), porosity, and chemical composition. In general, biochar produced at higher pyrolysis temperatures has more carbon, alkalinity, and stability, whereas biochar produced at lower temperatures is more porous and has higher nutrient availability [[29], [30], [31]]. The EC of BC generally increases with pyrolysis temperature due to enhanced carbonization and graphitic formation. This increase is not indefinite, studies indicate that optimum EC occurs between 600 and 800 °C, after which EC gains become marginal, suggesting a threshold for conductive BC production [32,33]. Reported inhibitory EC thresholds in AD range from 35 to 50 mS/cm for the liquid phase [34]; the EC of biochar itself is substantially higher, though its effect on reactor EC depends on dose, dilution, and solubilization rate. At 300, the EC of BC is 0.09, and it increased to 0.4 mS/cm at 700 °C [35]. Rice husks contain high amounts of silicon and mineral elements, which relate strongly with the ash content in the biochar [35,36].

The aim of this study was to inventigate the influence of rice husk biochar pyrolyzed at temperatures from 300 to 900 °C on the anaerobic digestion of food waste. Through the combination of comprehensive physicochemical characterization, biomethane production assays, and fermentation residue analysis, the study attempted to shed light on how pyrolysis temperature affects the functional characteristics of biochar and its performance as an additive in AD systems. This approach directly operationalises circular-economy principles by converting a problematic residue (rice husk) into an efficient additive that up-scales food waste valorisation in anaerobic digestion. Recent reviews emphasise that the addition of waste-derived biochar to AD can close material loops and stabilize the process, and also methane yields, and that therefore biochar-enhanced AD is a circular solution. While biochar additions to food-waste AD have been explored, including studies with rice-husk biochar, systematic mapping of a wide pyrolysis-temperature series (300-900 °C) combined with multi-modal characterization and digestate FT-IR to deliver structure–function relationships has been lacking; the present study fills this gap.

Section snippets

Biochar preparation and characterization

Rice husk was obtained from a local farm in San Jose Nueva Ecija, Philippines. The rice husk underwent size reduction using a stainless steel grinding machine and was passed through a mesh sieve (particle size <1.0 mm) [37]. The rice husks were dried in an oven dryer (WAMED, KBC-65W, Warsaw, Poland) at 105 °C for 24 h [22,38]. The processed rice husk was pyrolyzed using a muffle furnace (SNOL, January 8, 1100, Utena, Lithuania) at 300, 500, 700, and 900 °C for a 60-min retention time to produce

Physicochemical properties of rice husk biochars

The physical and chemical properties of biochar obtained at different temperatures are shown in Table 1. Data shows that pyrolysis temperature influenced TS, VS, AC, and FC. TS increased with temperature due to stronger carbonization and reduced moisture [29,37,54], while VS and VM declined as volatile compounds were lost. Ash content increased by ∼46% from BC300 to BC900, reflecting mineral concentration, and FC rose slightly (31.11 → 34.46%). Overall, higher temperatures decomposed the labile

Conclusion

This study revealed that pyrolysis temperature strongly shapes the biomethane yield and microbial dynamics in anaerobic digestion (AD) of food waste. Low-temperature rice husk biochar (BC300) achieved the highest biomethane production (335.76 mL/g-VS), which was closely linked to its porous structure, abundant functional groups, and nutrient-rich surface that favored microbial colonization. Microbial analysis showed that BC300 enriched methanogenic archaea and syntrophic bacteria, enhancing

CRediT authorship contribution statement

Marvin T. Valentin: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Katarzyna Ewa Kosiorowska:** Writing – review & editing, Visualization, Software, Resources, Methodology, Investigation, Conceptualization. **Agata Siedlecka:** Writing – review & editing, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Kacper**

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Declaration of competing interest

The authors declare no competing interest.